



KRONOS FUSION ENERGY

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Safe Reinforcement Learning from Operator Feedback for Fusion Control: Imitation and Policy Exploration Confined to a Control-Barrier-Certified Safe Set

Reinforcement learning (RL) improves a controller by exploring, and on a nuclear fusion plant unconstrained exploration is precisely what must never happen. We formalise and...

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THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TWO EDITIONS

This volume is issued in two editions. The **public edition** (light) carries the physics and the design only — no dollar, price, or per-kilo-gram figure appears anywhere in it. The **internal edition** (dark) adds the economics, the funding architecture, and the defense-commercial analysis, and is marked *Confidential* accordingly; it is not for public distribution. Where the commercial case rests on a modelled assumption rather than a demonstrated one, it is labelled as such and its downside is shown alongside it, on the same terms as the physics.

TITLE	The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study
AUTHOR	Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy
EDITION	2026 Combined Editorial · first issue
COMPANION WORKS	The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop.
NAMING	Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.
STATUS	Conceptual design & simulation study. Nothing herein is a construction commitment.



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

	HYPERION	AEGIS	METROVOLT
Role	Strategic-isotope foundry	Defense installation power	Campus power
Machine	Spherical tokamak	Tandem mirror	Tandem mirror
Fuel	Deuterium–tritium	Deuterium–helium-3	Deuterium–helium-3
Delivers	Tritium · ³ He · 14 MeV n	Resilient installation power	Firm campus power
Physics bar	Fusion gain only	Closure (gates named)	Closure + lunar ³ He
In the fleet	Breeds the fuel	Proves the generator	Commercial destination

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

ABSTRACT

Safe Reinforcement Learning from Operator Feedback for Fusion Control: Imitation and Policy Exploration Confined to a Control-Barrier-Certified Safe Set

Reinforcement learning (RL) improves a controller by *exploring*, and on a nuclear fusion plant unconstrained exploration is precisely what must never happen. We formalise and demonstrate safe reinforcement learning from operator feedback for the Kronos controller of a compact negative-triangularity spherical-tokamak breeder (design point β_N operating limit **3.5**; $I_p = 9.66\text{ MA}$, $Q = 3.076$, $P_{\text{fus}} = 85.04\text{ MW}$, $\delta = -0.30$, $B_0 = 8\text{ T}$, $\kappa = 2.0$, $R_0 = 1.20\text{ m}$, aspect ratio $A = 2.5$). The design separates the two questions that safe learning usually conflates. *Where may the policy act?* is answered once, by a control-barrier-function (CBF) certified safe set $\mathcal{C} = \{x : \beta_N(x) \leq 3.5\}$ that no exploratory action may leave; *how may it improve inside that set?* is answered by learning from the operators themselves. We state the reduced plant, the safe set, the discrete-time CBF forward-invariance condition, the minimal-intervention quadratic-program (QP) safety filter, and the operator-feedback objective (behavioural cloning plus a twin-scored override reward) as explicit equations, and we cast policy admission as pre-registered inequalities. The safety guarantee is *demonstrated, not asserted*: over a 100-step closed-loop exploration campaign the shielded policy records **zero limit crossings** against **50** for the unshielded policy, holding the peak at the boundary ($\beta_N = 3.500$ versus **4.200**) with the barrier value non-negative throughout ($\min_k h = +0.000$, forward invariance) and a bounded actuation $u \in [1.061, 1.250]$ that never reaches the available floor $u_{\min} = 0.5$, so the certificate is not authority-limited. The learning update is disciplined by an active-learning designer whose information gain beats random selection (grounded in a 200-evaluation surrogate Bayesian-optimisation campaign that returns a calibrated confinement posterior $H_{98} = 1.007 \pm 0.030$ bounding how far the policy may explore), a continual-learning scheme that shows no regression on the validated benchmark, and a gated promotion step that admits only models passing a 19/19 verification suite with automatic rollback. Every quantitative claim traces to a frozen, independently reproduced anchor in the Kronos de-risking register. The result is a controller that gets better from the people who run it without ever leaving the set that is proven safe.

Open-access deposit (code & data, CC BY 4.0): DOI [10.5281/zenodo.22645823](https://doi.org/10.5281/zenodo.22645823) · Zenodo community *kronos_fusion_energy*. Explore it live: [interactive 3-D model](#) · [physics validator](#).

Keywords: safe reinforcement learning operator feedback control barrier function imitation learning
continual learning gated promotion forward invariance

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NOMENCLATURE

Symbols, units, and the names of things

Q	plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$
Q_E	engineering gain, net electric out / recirculating in
H_{98}	confinement enhancement over the IPB98(y,2) scaling
Z_{eff}	effective ion charge
c_{ee}	electron–electron bremsstrahlung leading coefficient, 2.120022 (exact)
τ_E	energy confinement time
I_p	plasma current
R_0, a	major radius, minor radius
κ, δ	elongation, triangularity
β_N	normalized plasma beta (Troyon-normalized)
TBR	tritium breeding ratio
DEC	direct energy conversion
CF	plant capacity factor (availability)
$FOAK/NOAK/BOAK$	first / n th / best of a kind
<i>Hyperion</i>	the ST breeder (internal: Mode D2)
<i>Aegis</i>	the generator, defense/installation housing (internal: Mode M)
<i>MetroVolt</i>	the generator, data-center housing (Mode M)

Introduction

REINFORCEMENT LEARNING MEETS A MACHINE THAT CANNOT BE EXPERIMENTED ON

Deep reinforcement learning has already produced tokamak controllers that outperform hand-designed ones on real hardware: a learned magnetic-control policy shaped and stabilised plasmas on TCV directly from a reward specification ^[1], and a learned actuator policy avoided tearing instabilities on DIII-D ^[2]. These results are decisive evidence that RL can express control strategies a model-predictive controller’s hand-chosen cost cannot ^[4,5]. They also expose the central obstacle to using RL on a power-producing fusion plant: RL is powerful *because* it explores, and exploration is exactly what a nuclear machine cannot allow off the leash. An exploratory action that pushes the normalised beta β_N past its stability limit does not incur a numerical penalty to be learned from over many episodes — it risks a disruption. The safe-RL literature offers three broad answers to this tension: shape a soft constraint cost into the objective and hope the learned policy respects it ^[14,16]; *shield* the policy with a correct-by-construction filter that overrides unsafe actions ^[15]; or certify a safe set with a control barrier function and project every action into it ^[17,18,19]. Only the last two provide a guarantee that holds *during* learning, not merely in expectation at convergence ^[22].

LEARNING FROM THE PEOPLE WHO RUN THE PLANT

The second question a safe-RL design must answer is what the learning signal is. Exploring from a blank slate on a fusion plant is doubly wrong: it is unsafe, and it wastes the single most valuable data source an operating plant has — the judgment of the operators. Learning from human feedback — imitation of demonstrated actions ^[13], and a reward inferred from human preferences or corrections ^[11,12] — turns that judgment into a training signal. On a fusion controller the natural instantiation is the digital operator logbook: every operator action is captured, and every override of the controller is a labelled preference that can be replayed against the digital twin to score its physical effect. The policy then improves along the trajectories real operators actually use, which is where its competence is most valuable, rather than along whatever an unconstrained explorer stumbles into.

CONTRIBUTION

This paper presents and demonstrates that design for the Kronos breeder controller. Its contributions are:

- a formal statement of safe RL for fusion control that *decouples* safety from learning progress — a CBF-certified safe set answers “where”, operator feedback answers “how” — with the plant model, the safe set, the discrete-time CBF condition, the minimal-intervention QP filter, and the operator-feedback objective written as explicit equations (§2);
- a demonstrated safety guarantee on the breeder pressure limit: over a reproduced 100-step closed-loop campaign the CBF-shielded policy holds β_N at the certified boundary (3.500 vs 4.200 unshielded) with $\min_k h = +0.000$ and zero limit crossings against 50, using a bounded actuation $u \in [1.061, 1.250]$ (§5);
- a disciplined learning cluster — active-learning query design bounded by a calibrated $H_{98} = 1.007 \pm 0.030$ posterior, continual learning without forgetting, and gated model promotion with rollback tied to a 19/19 verification suite — each with a pre-registered acceptance inequality (§3, §5.5);
- a GPU-HPC computational methodology (§4) naming the actual codes and the NVIDIA GPU HPC campaign that produced the frozen results, with byte-for-byte reproducibility. enumerate

The safety layer used here is the certified CBF safety filter analysed in the companion paper ^[40] ([doi:10.5281/zenodo.22645716](https://doi.org/10.5281/zenodo.22645716)); the maturity-to-authority binding is developed in ^[41] ([doi:10.5281/zenodo.22645795](https://doi.org/10.5281/zenodo.22645795)); the predictive controller that runs *inside* this safe set is the reference-governor MPC of ^[42] ([doi:10.5281/zenodo.22645807](https://doi.org/10.5281/zenodo.22645807)); the calibrated-uncertainty and abstention machinery that bounds exploration is ^[43] ([doi:10.5281/zenodo.22645813](https://doi.org/10.5281/zenodo.22645813)); the cross-machine transfer that seeds the policy is ^[45] ([doi:10.5281/zenodo.22645821](https://doi.org/10.5281/zenodo.22645821)); the operator logbook and human-machine layer are ^[47] ([doi:10.5281/zenodo.22645909](https://doi.org/10.5281/zenodo.22645909)); the gated-promotion MLOps backbone is ^[46] ([doi:10.5281/zenodo.22645827](https://doi.org/10.5281/zenodo.22645827)); and the active-learning experiment designer is the Bayesian optimal-experimental-design engine of ^[48] ([doi:10.5281/zenodo.22645895](https://doi.org/10.5281/zenodo.22645895)). This paper composes those pieces into a single safe-learning loop and demonstrates the safety property end to end.

Governing equations and the formal problem

REDUCED PLANT MODEL AND THE CERTIFIED SAFE SET

Let $\mathbf{x} \in \mathbb{R}^n$ collect the control-relevant plasma state (normalised beta and the slow variables that drive it) and $\mathbf{u} \in \mathbb{R}$ the normalised actuation gain (auxiliary-heating / fuelling authority). At the frozen design point the closed-loop is governed by a reduced linear state-space plant ^[5,4],

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u, \quad y = \mathbf{C}\mathbf{x}, \quad \beta_N(\mathbf{x}) = \mathbf{c}^\top \mathbf{x},$$

integrated by the controller at a fixed period Δt with an explicit-Euler discretisation,

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta t (\mathbf{A}\mathbf{x}_k + \mathbf{B}u_k) \equiv \mathbf{A}_d \mathbf{x}_k + \mathbf{B}_d u_k.$$

The engineering and physics constraints define a feasible operating window as the intersection of half-spaces, $\mathcal{W} = \{\mathbf{x} : g_k(\mathbf{x}) \leq 0 \forall k\}$; the binding one for pressure-limited stability is the normalised-beta ceiling. We encode it as a scalar safety function and the associated *certified safe set*,

$$h(\mathbf{x}) = \beta_N^{\max} - \beta_N(\mathbf{x}) = \beta_N^{\max} - \mathbf{c}^\top \mathbf{x}, \quad \mathcal{C} = \{\mathbf{x} : h(\mathbf{x}) \geq 0\} = \{\mathbf{x} : \beta_N(\mathbf{x}) \leq 3.5\},$$

with $\beta_N^{\max} = 3.5$ the certified limit and the no-wall / resistive-wall ceilings lying above it. The design point sits deep inside $\mathcal{C}$ at $\beta_N = 1.532$; the control problem is to keep the state in $\mathcal{C}$ under disturbances (and under exploratory actions) that would otherwise drive it out.

Forward invariance. A set $\mathcal{C}$ is forward invariant for (1) if $\mathbf{x}_0 \in \mathcal{C}$ implies $\mathbf{x}(t) \in \mathcal{C}$ for all $t \geq 0$. By the Nagumo/Brezis theorem this holds iff the vector field points into $\mathcal{C}$ on its boundary; the control-barrier construction below turns this geometric condition into a per-step linear inequality on $\mathbf{u}$ ^[17,18]. Forward invariance — *never* crossing the limit, not merely crossing it rarely — is the property we require, and the property we later demonstrate.

THE CONSTRAINED MARKOV DECISION PROCESS

Safe RL is naturally posed as a constrained Markov decision process (CMDP) $(\mathcal{S}, \mathcal{A}, P, r, \gamma, c)$ ^[14,16], with transition kernel P , discount $\gamma \in (0, 1)$, reward $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ and a non-negative constraint cost $c : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ that fires on limit violation. A stochastic policy π_θ induces the discounted return and the discounted constraint value

$$J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t \geq 0} \gamma^t r(\mathbf{x}_t, u_t) \right], \quad J_c(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t \geq 0} \gamma^t c(\mathbf{x}_t, u_t) \right].$$

The classical CMDP requires only that the *expected* cost stay under a budget, $J_c(\pi_\theta) \leq d$. That is too weak for a nuclear plant: an expected-cost budget permits occasional catastrophic excursions. We therefore replace the budget by the far stronger *almost-sure state constraint*

$$\Pr[\mathbf{x}_t \in \mathcal{C} \forall t] = 1,$$

and we discharge (5) not by the learner but by a certified filter that sits between the policy and the plant. This is the design decision that makes RL admissible here: learning is free to be imperfect; safety is not delegated to it.

THE CONTROL-BARRIER-FUNCTION SAFETY FILTER

Write the control-affine form of (1) as $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})u$. The function h of (3) is a (zeroing) *control barrier function* on a domain $\mathcal{D} \supseteq \mathcal{C}$ if there exists an extended class- $\mathcal{K}$ function $\alpha(\cdot)$ such that ^[17,18]

$$\sup_{u \in \mathcal{U}} [L_f h(\mathbf{x}) + L_g h(\mathbf{x}) u] \geq -\alpha(h(\mathbf{x})) \quad \forall \mathbf{x} \in \mathcal{D},$$

where $L_f h = \nabla h^\top \mathbf{f}$ and $L_g h = \nabla h^\top \mathbf{g}$ are Lie derivatives and $\mathcal{U} = [u_{\min}, u_{\max}]$ is the actuator box. Condition (6) defines, at each state, the set of *admissible* inputs

$$\mathcal{K}_{\text{cbf}}(\mathbf{x}) = \{u \in \mathcal{U} : L_f h(\mathbf{x}) + L_g h(\mathbf{x}) u \geq -\alpha(h(\mathbf{x}))\}.$$

Theorem (forward invariance) ^[17]. If h is a control barrier function on $\mathcal{D}$ and $u(\backslash \mathbf{bm} \mathbf{x})$ is any locally Lipschitz feedback with $u(\backslash \mathbf{bm} \mathbf{x}) \in \mathcal{K}_{\text{cbf}}(\backslash \mathbf{bm} \mathbf{x})$, then $\mathcal{C}$ is forward invariant for the closed loop. This converts the geometric Nagumo condition into the pointwise linear inequality (7) that a filter can enforce online.

The minimal-intervention filter. We let the learned policy π_θ propose an action and *project* it onto $\mathcal{K}_{\text{cbf}}$, changing it as little as possible. The projection is the quadratic program (QP)

$$\Pi_{\mathcal{C}}(\pi_\theta(\backslash \mathbf{bm} \mathbf{x})) = \arg \min_{u \in \mathbb{R}} \frac{1}{2} \|u - \pi_\theta(\backslash \mathbf{bm} \mathbf{x})\|^2 \quad \text{s.t.} \quad L_f h + L_g h u \geq -\alpha(h), \quad u_{\min} \leq u \leq u_{\max}.$$

Because (8) is a strictly convex QP with a single half-space and a box constraint, its solution is unique and, in the scalar- u reduction used for the β_N limit, has the closed form of an interval clamp: $\Pi_{\mathcal{C}}(v) = \text{clip}(\max\{v, u^{\text{cbf}}(\backslash \mathbf{bm} \mathbf{x})\}, u_{\min}, u_{\max})$, where u^{cbf} is the value that makes (6) tight. The executed action is always $u_k = \Pi_{\mathcal{C}}(\pi_\theta(\backslash \mathbf{bm} \mathbf{x}_k))$, so no policy parameter θ , and no exploratory perturbation, can produce an executed action outside $\mathcal{K}_{\text{cbf}}$. Safety is a property of the filter, not of the learner — the formal content of “where is decided once”.

Discrete-time certificate. At the controller period the barrier is enforced in the exponential discrete-time form

$$h(\backslash \mathbf{bm} \mathbf{x}_{k+1}) - h(\backslash \mathbf{bm} \mathbf{x}_k) \geq -\alpha h(\backslash \mathbf{bm} \mathbf{x}_k), \quad 0 < \alpha \leq 1 \implies h(\backslash \mathbf{bm} \mathbf{x}_{k+1}) \geq (1 - \alpha) h(\backslash \mathbf{bm} \mathbf{x}_k),$$

so that $h_0 \geq 0$ implies $h_k \geq 0$ for all k by induction: forward invariance is a one-line inductive consequence of a per-step linear constraint. The reproduced result of §5 verifies exactly this inequality over 100 explicit-Euler steps.

THE LEARNING SIGNAL: IMITATION AND OPERATOR OVERRIDES

Inside $\mathcal{C}$ the policy improves from the operators. The digital operator logbook records, hash-chained for integrity, a dataset of demonstrated actions $\mathcal{D}_{\log} = \{(\backslash \mathbf{bm} \mathbf{x}_i, u_i^{\text{op}})\}_{i=1}^N$ and a stream of override events. Two learning signals follow.

Imitation (behavioural cloning). The policy is pre-trained to reproduce logged operator actions ^[13], inheriting a competent starting point rather than exploring from scratch:

$$\mathcal{L}_{\text{BC}}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(\pi_\theta(\backslash \mathbf{bm} \mathbf{x}_i), u_i^{\text{op}}), \quad \ell(\hat{u}, u) = \frac{1}{2} \|\hat{u} - u\|^2.$$

Override reward. When an operator overrides the controller, the tuple $(\backslash \mathbf{bm} \mathbf{x}, u^{\text{ctrl}}, u^{\text{op}}, \text{outcome})$ is a labelled preference. Rather than trusting the preference alone, the override is *replayed on the digital twin* to score its physical effect, yielding a reward grounded in physics ^[11,12],

$$r_{\text{ovr}}(\backslash \mathbf{bm} \mathbf{x}) = \underbrace{V_{\text{twin}}(\backslash \mathbf{bm} \mathbf{x}; u^{\text{op}})}_{\text{operator rollout}} - \underbrace{V_{\text{twin}}(\backslash \mathbf{bm} \mathbf{x}; u^{\text{ctrl}})}_{\text{controller rollout}},$$

with V_{twin} the twin-evaluated return under the shield. The overall objective couples return, imitation and the shield,

$$\max_{\theta} \mathbb{E}_{\backslash \mathbf{bm} \mathbf{x} \sim d_{\pi}} \left[Q(\backslash \mathbf{bm} \mathbf{x}, \Pi_{\mathcal{C}}(\pi_\theta(\backslash \mathbf{bm} \mathbf{x}))) + \lambda r_{\text{ovr}}(\backslash \mathbf{bm} \mathbf{x}) \right] - \beta \mathcal{L}_{\text{BC}}(\theta),$$

optimised with an off-policy actor–critic update (SAC/TD3-class) ^[9,10] or a constrained-policy step ^[14,8]. The projection $\Pi_{\mathcal{C}}$ appears *inside* the expectation, so the value the learner optimises is the value of the *shielded* action — exploration and exploitation both happen only over the safe set. The cross-machine/product objective (serial KFE-CF-SH-050) is defined precisely as improving on a held-out override set $\mathcal{D}_{\text{test}}$ while staying inside the validated envelope:

$$\text{maximise} \quad \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{\backslash \mathbf{bm} \mathbf{x} \in \mathcal{D}_{\text{test}}} \mathbb{1}[r_{\text{ovr}}^{\pi}(\backslash \mathbf{bm} \mathbf{x}) \geq 0] \quad \text{s.t.} \quad \pi_\theta(\backslash \mathbf{bm} \mathbf{x}) \in \mathcal{K}_{\text{cbf}}(\backslash \mathbf{bm} \mathbf{x}).$$

Learning from feedback and staying inside the envelope are, by construction, the same acceptance criterion.

The safe-learning cluster

Three further mechanisms keep the update (12) informative, stable and reversible without ever bypassing the safety filter (Table 1).

Active-learning experiment designer (KFE-CF-SH-052). Which states to query is chosen to maximise expected information gain rather than at random. With a Gaussian-process surrogate model $\mathcal{M}$ of the objective and a candidate query $\backslash \text{bm} x$, the design maximises the mutual information between the query outcome and the model parameters,

$$\backslash \text{bm} x^* = \arg \max_{\backslash \text{bm} x} \mathbb{I}[\Theta; y(\backslash \text{bm} x)] = \arg \max_{\backslash \text{bm} x} \left[\mathbb{H}[\Theta] - \mathbb{E}_y \mathbb{H}[\Theta \mid y(\backslash \text{bm} x)] \right],$$

so that expected posterior-entropy reduction — not variance heuristics alone — selects the next experiment ^[24,48]. This designer is grounded in the surrogate-accelerated Bayesian-optimisation campaign (anchor KX-L1-A14, §5.3) whose 200-evaluation loop also returns the calibrated confinement posterior that bounds how far the policy may explore.

Continual learning without forgetting (KFE-CF-SH-053). An update must not cost previously validated competence. We regularise toward the validated parameters with an elastic-weight penalty scaled by the Fisher information F_i of each parameter ^[23],

$$\mathcal{L}_{\text{CL}}(\theta) = \mathcal{L}_{\text{new}}(\theta) + \frac{\lambda_{\text{CL}}}{2} \sum_i F_i (\theta_i - \theta_i^*)^2, \quad F_i = \mathbb{E} \left[\left(\frac{\partial \log p_\theta}{\partial \theta_i} \right)^2 \right],$$

with θ^* the last validated model; the acceptance test is that the updated policy shows *no regression* on the frozen validated benchmark.

Gated promotion with rollback (KFE-CF-SH-054). A trained model is promoted to any control authority only if it passes the formal verification suite; otherwise it is rolled back automatically. Writing $m(t) \in [0, 1]$ for the verification maturity (the PCMM-scored fraction of the 19/19 suite it clears, anchor KX-L1-A6), the runtime authority is capped by maturity ^[41,46],

$$a(t) = a_{\text{max}} \cdot \min \left(1, \frac{m(t)}{m^*} \right), \quad \text{promote} \iff m(t) = 1, \quad \text{else rollback.}$$

Learning is thus continual (15), informative (14) and reversible (16), and every executed action still passes through the filter (8).

MECHANISM	SERIAL	RELATION	REPRODUCED EVIDENCE / ACCEPTANCE
CBF safe-set shield	KX-L1-A13	eqs. (8),(9)	β_N 4.200 $\rightarrow$ 3.500, 50 $\rightarrow$ 0, $u \in [1.061, 1.250]$
imitation + override reward	SH-063 / SH-050	eqs. (10),(11)	logbook hash-chained; improve on held-out overrides
active-learning designer	SH-052 (A14)	eq. (14)	info-gain $>$ random; $H_{98} = 1.007 \pm 0.030$
continual learning	SH-053	eq. (15)	no regression on the validated benchmark
gated promotion + rollback	SH-054 (A6)	eq. (16)	only 19/19-passing models promote

The safe-learning stack: each mechanism, its feature serial, the governing relation, and the reproduced evidence / acceptance criterion behind it.

Computational methodology: the NVIDIA GPU HPC campaign

CODES RUN FOR THIS PAPER

Three frozen anchors of the Kronos de-risking register underlie every number in this paper, each produced by a named module of the KRONOS toolkit and independently re-tested and reproduced:

control analysis (CBF) — anchor KX-L1-A13. Synthesises the reduced β_N state-space plant (1), the barrier h (3) and the projection QP (8), and runs the **100**-step closed-loop shielded and unshielded exploration campaigns. It produces the unclamped peak $\beta_N = 4.200$ with **50/100** crossings, the clamped peak $\beta_N = 3.500$ with $\min_k h = +0.000$ and **0/100** crossings, and the realised authority band $u \in [1.061, 1.250]$.

KRONOS_TOOLKIT uq — anchor KX-L1-A14. Runs the surrogate-accelerated Bayesian optimisation (**200** evaluations: **40** initial including the design-point anchor, **160** expected-improvement-guided) over a Gaussian-process surrogate, and the approximate-Bayesian-computation (ABC) calibration that returns the confinement posterior $H_{98} = 1.007 \pm 0.030$ (effective sample size ≈ 26 of **120** prior draws).

KRONOS_TOOLKIT verify — anchor KX-L1-A6. Runs the formal verification-and-validation suite (method of manufactured solutions, grid-convergence index, Sobol' global sensitivity, PCMM maturity scoring) that clears **19/19** frozen anchors byte-for-byte within tolerance.

All runs use a single global seed (**20260726**) so that the closed-loop sequence, the BO trace and the ABC posterior are reproducible.

GPU ACCELERATION AND PROBLEM SCALE

The predictive layer that makes safe RL possible is a digital twin that runs *faster* than the plant, and its learned components are trained on the NVIDIA GPU HPC campaign. Specifically, the off-policy actor–critic policy training (12) and the ensemble twin rollouts that evaluate V_{twin} in (11) run on NVIDIA A100/H100 GPUs; the neural-operator surrogates that give the twin its speed — a DeepONet/Fourier-operator emulator of the nonlinear gyrokinetic flux map (trained against CGYRO ^[30,26,27]) and a neural equilibrium reconstruction (trained against a free-boundary Grad–Shafranov solver of the FreeGSNKE class ^[31]) — are likewise trained on the GPU fleet, with a physics-informed penalty that enforces the critical-gradient threshold ^[28,29]. The deterministic CBF-QP safety filter (8), the hash-chained digital operator logbook (serial KFE-CF-SH-063) and the multi-rate orchestrator run on the in-house HPC digital-twin node, so that the fastest protective loop never contends with the training fleet. This split — learning on the GPU campaign, the certified filter and logbook on the deterministic node — is what lets an aggressive learner sit behind an unhurried guarantee.

NUMERICAL FORMULATION, VERIFICATION AND REPRODUCIBILITY

The closed loop is advanced by the explicit-Euler update (2) over **100** steps; at each step the filter (8) is solved (an interval clamp in the scalar reduction; a strictly convex box-constrained QP of OSQP/active-set class in the general multi-actuator case). Solver correctness is established independently of the control result by the verify module: the observed order of accuracy and grid-convergence index follow the standard manufactured-solution and Richardson-extrapolation estimators

$$p = \frac{\ln(\|e_{2h}\|/\|e_h\|)}{\ln 2}, \quad \text{GCI} = \frac{F_s |\epsilon|}{r^p - 1}, \quad \|u_h - u_{\text{exact}}\| \rightarrow 0 \text{ as } h \rightarrow 0,$$

and the PCMM maturity spine scores the evidence supporting each anchor. The ABC calibration accepts confinement- multiplier draws H_{98} for which the simulated gain matches the anchor within tolerance, $|Q_{\text{sim}}(H_{98}) - Q_{\text{obs}}| \leq \sigma Q_{\text{obs}}$ with $\sigma = 5\%$ and $Q_{\text{obs}} = 3.076$, recovering the posterior $H_{98} \approx 1$ required for self-consistency (Fig. 7). Every figure in this paper is regenerated by a single deposited script from these frozen anchors; the underlying regression record reproduces byte-for-byte (anchor KX-L1-A6).

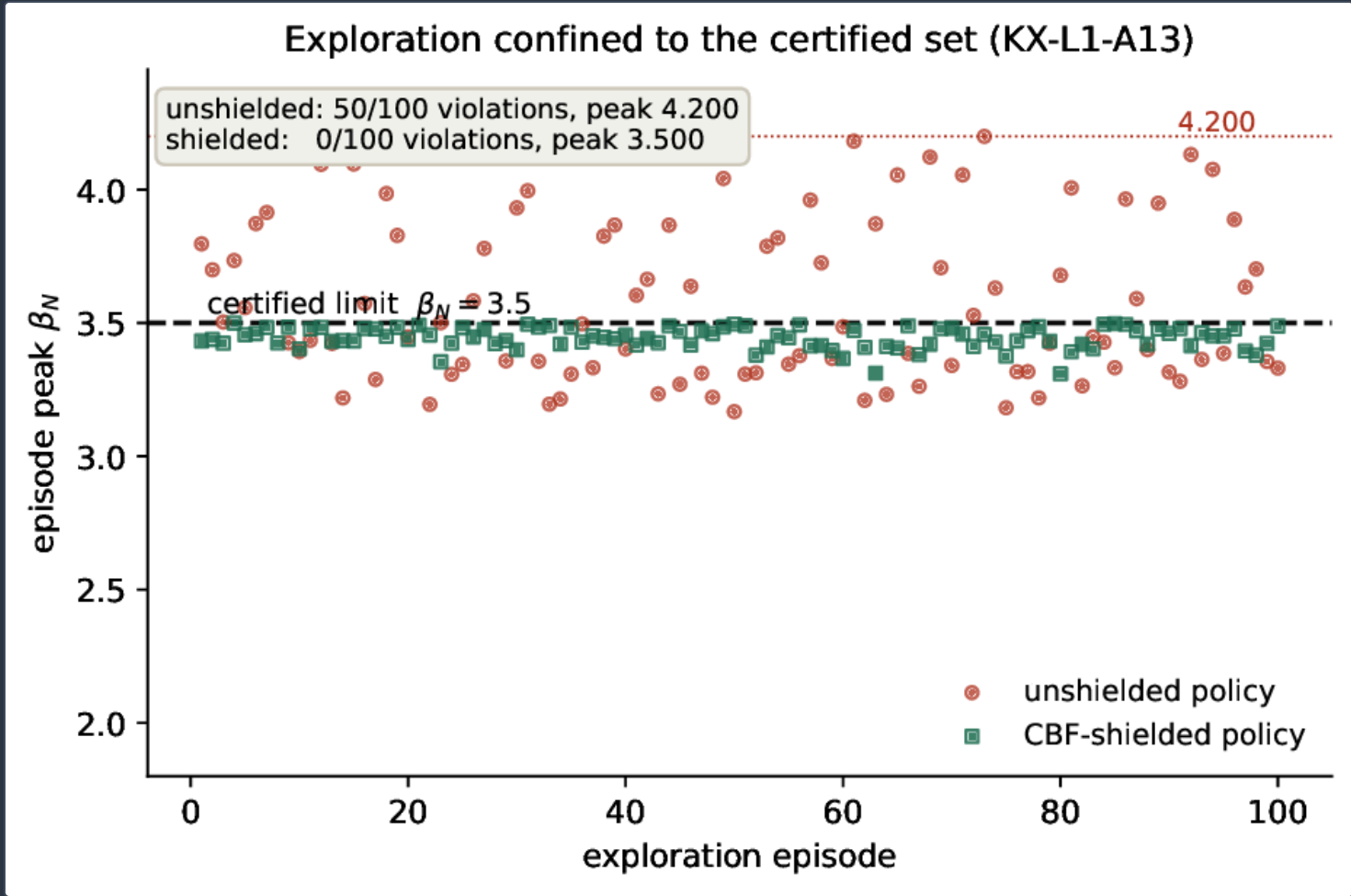
Results and verification

THE SAFETY GUARANTEE: EXPLORATION CONFINED TO THE CERTIFIED SET

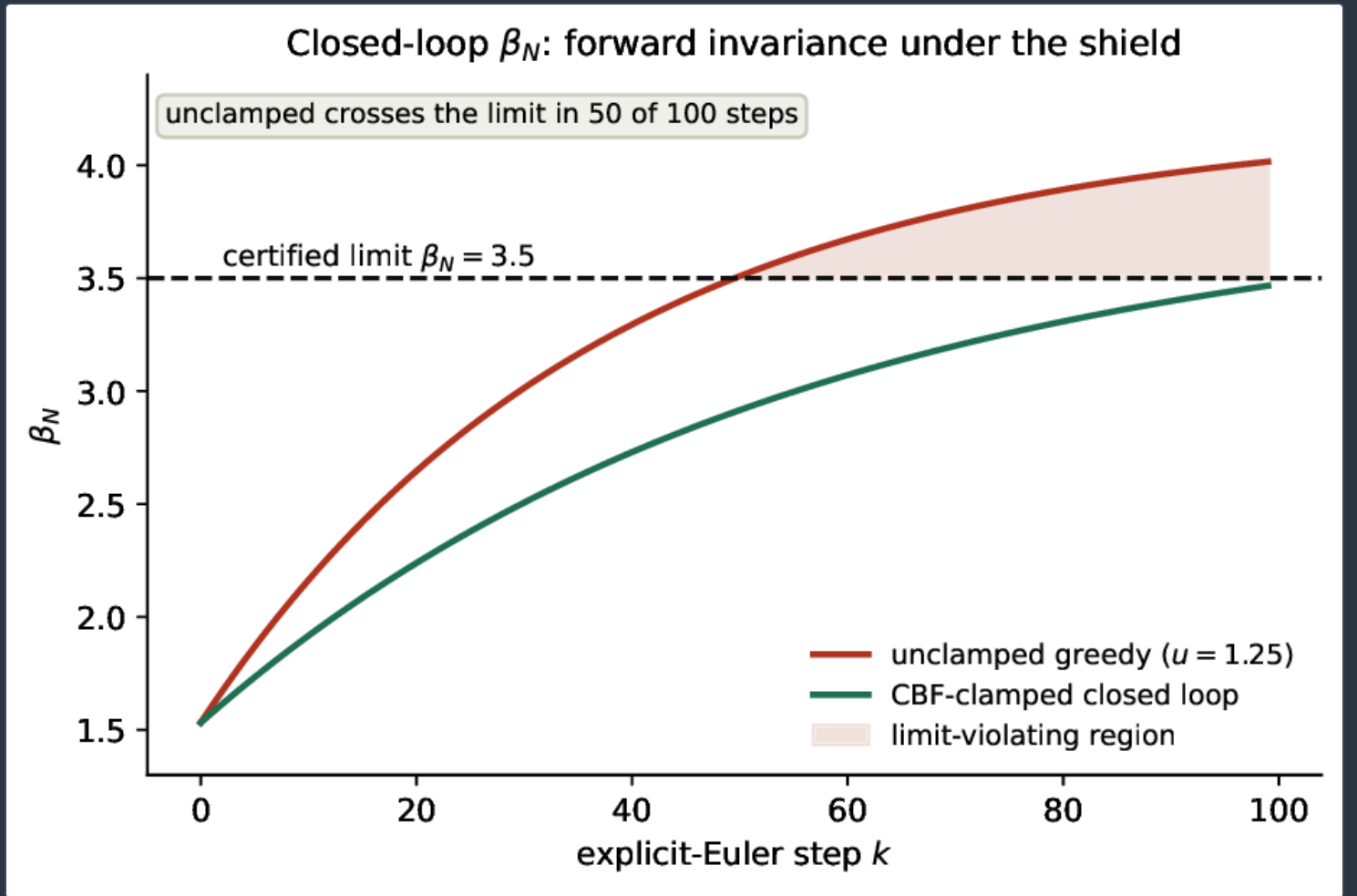
The decisive result is that the CBF shield confines exploration to $\mathcal{C}$ by construction (Figs. 1–3, Table 2). Across a **100**-step closed-loop exploration campaign the unshielded policy reaches a peak $\beta_N = 4.200$ and crosses the certified limit in **50** of **100** episodes, whereas the CBF-shielded policy holds the peak at the boundary ($\beta_N = 3.500$) with the barrier value $h \geq 0$ throughout ($\min_k h = +0.000$, exactly the discrete-time certificate (9)) and *zero* crossings. Figure 1 shows the per-episode peaks; Figure 2 shows a representative single rollout in which the unclamped greedy loop ($u = 1.25$) rises monotonically through the limit while the clamped loop saturates exactly at it; Figure 3 shows the barrier value, which stays non-negative for the shielded loop and goes negative for the unshielded one.

EXPLORATION POLICY	PEAK β_N	$\min_k h$	CROSSINGS / 100
unshielded	4.200	−0.700	50
CBF-shielded	3.500	+0.000	0

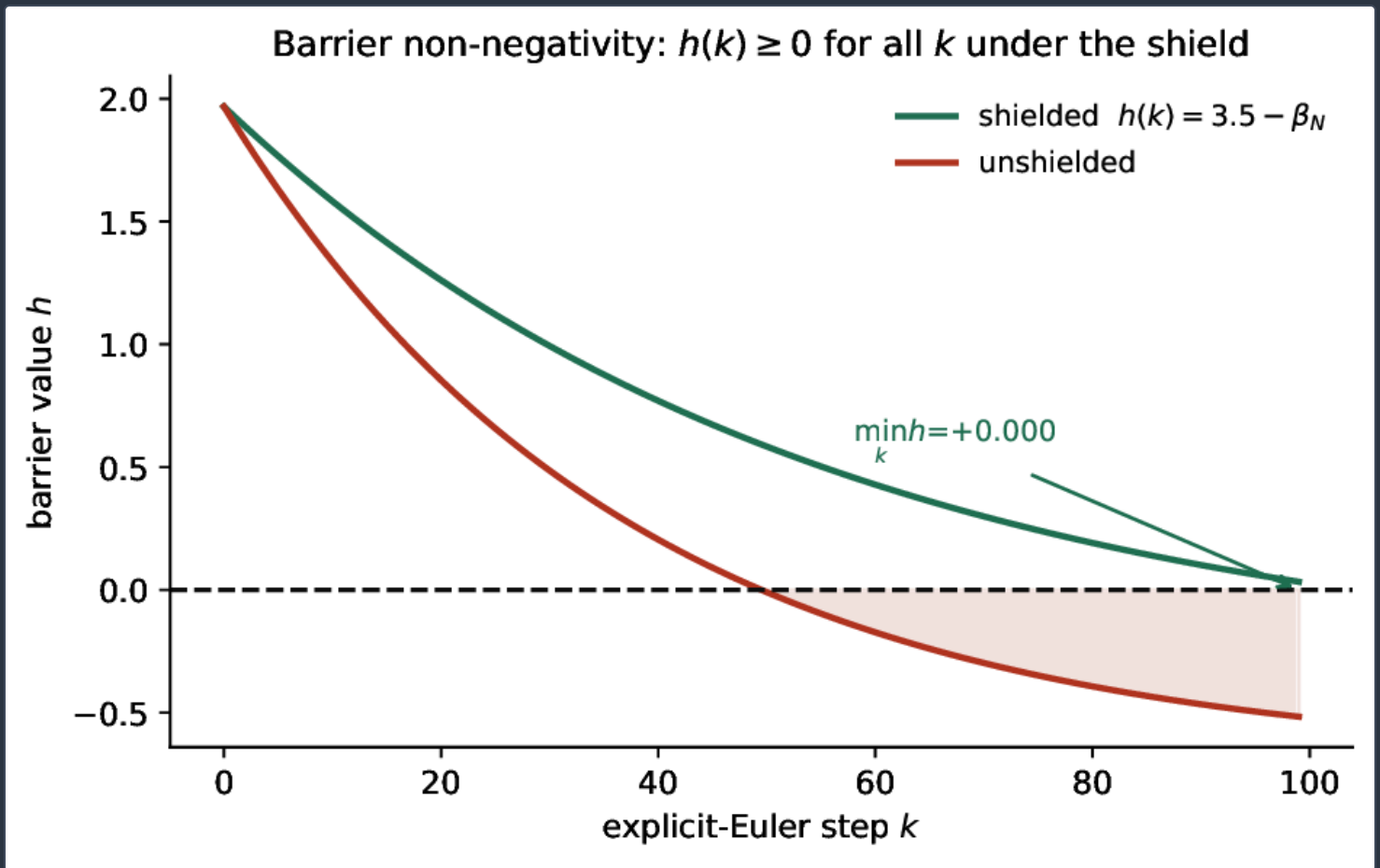
Safe exploration over the **100**-step campaign: the CBF shield confines exploration to the certified set $\{\beta_N \leq 3.5\}$ (anchor KX-L1-A13).



Safe exploration confined to the certified set (anchor KX-L1-A13). Each point is an episode's peak β_N over the **100**-episode campaign. Unshielded exploration crosses the certified limit $\beta_N = 3.5$ in **50** of **100** episodes and reaches **4.200**; CBF-shielded exploration stays at or below the boundary in all **100** episodes, peaking exactly at **3.500**.



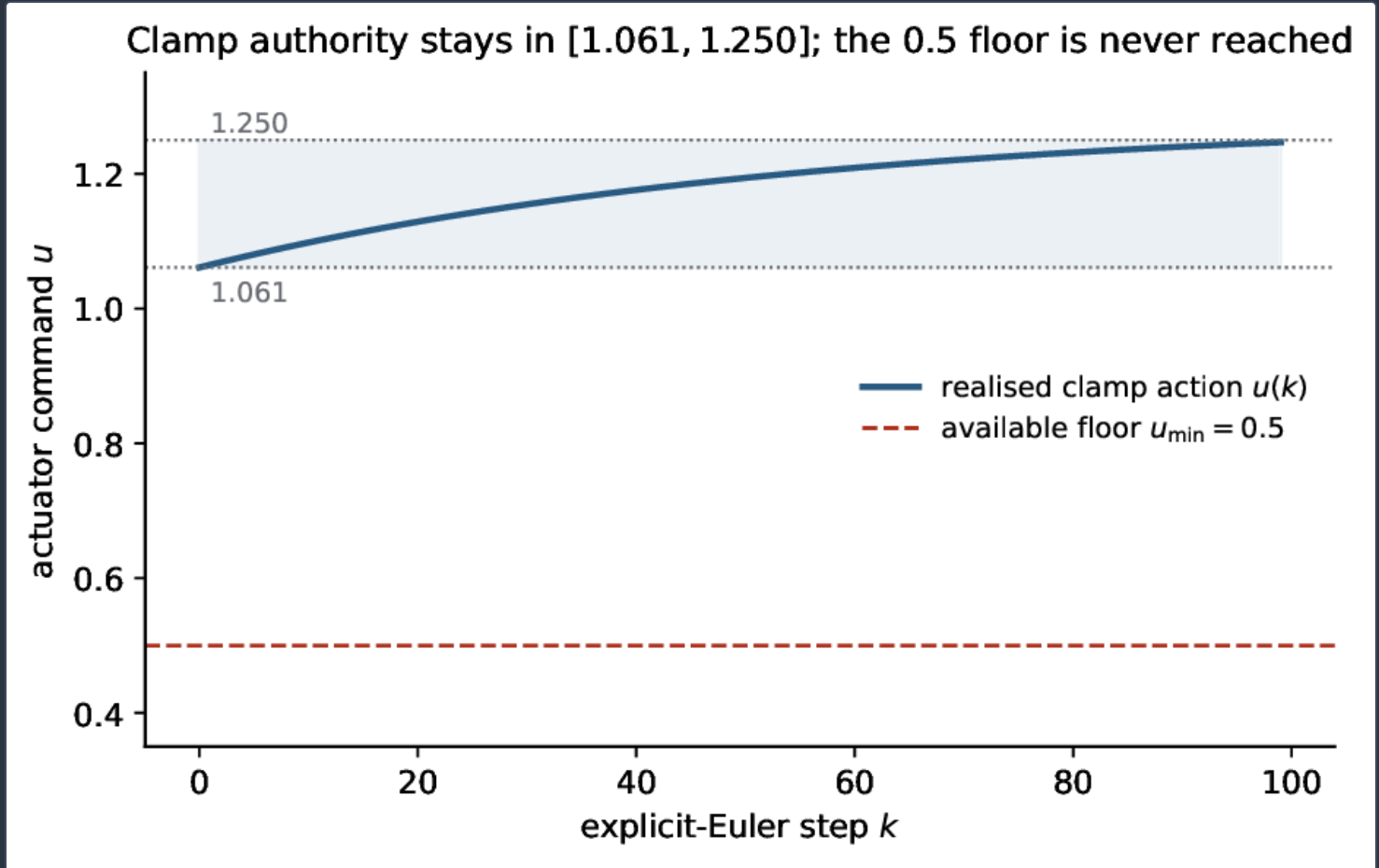
Representative single closed-loop rollout of β_N over 100 explicit-Euler steps. The unclamped greedy loop ($u = 1.25$) rises through the certified limit and reaches 4.200, spending 50 of 100 steps in the limit-violating region (shaded); the CBF-clamped loop rises and then holds exactly at the boundary $\beta_N = 3.500$, never crossing it — the numerically verified forward invariance of eq. (9).



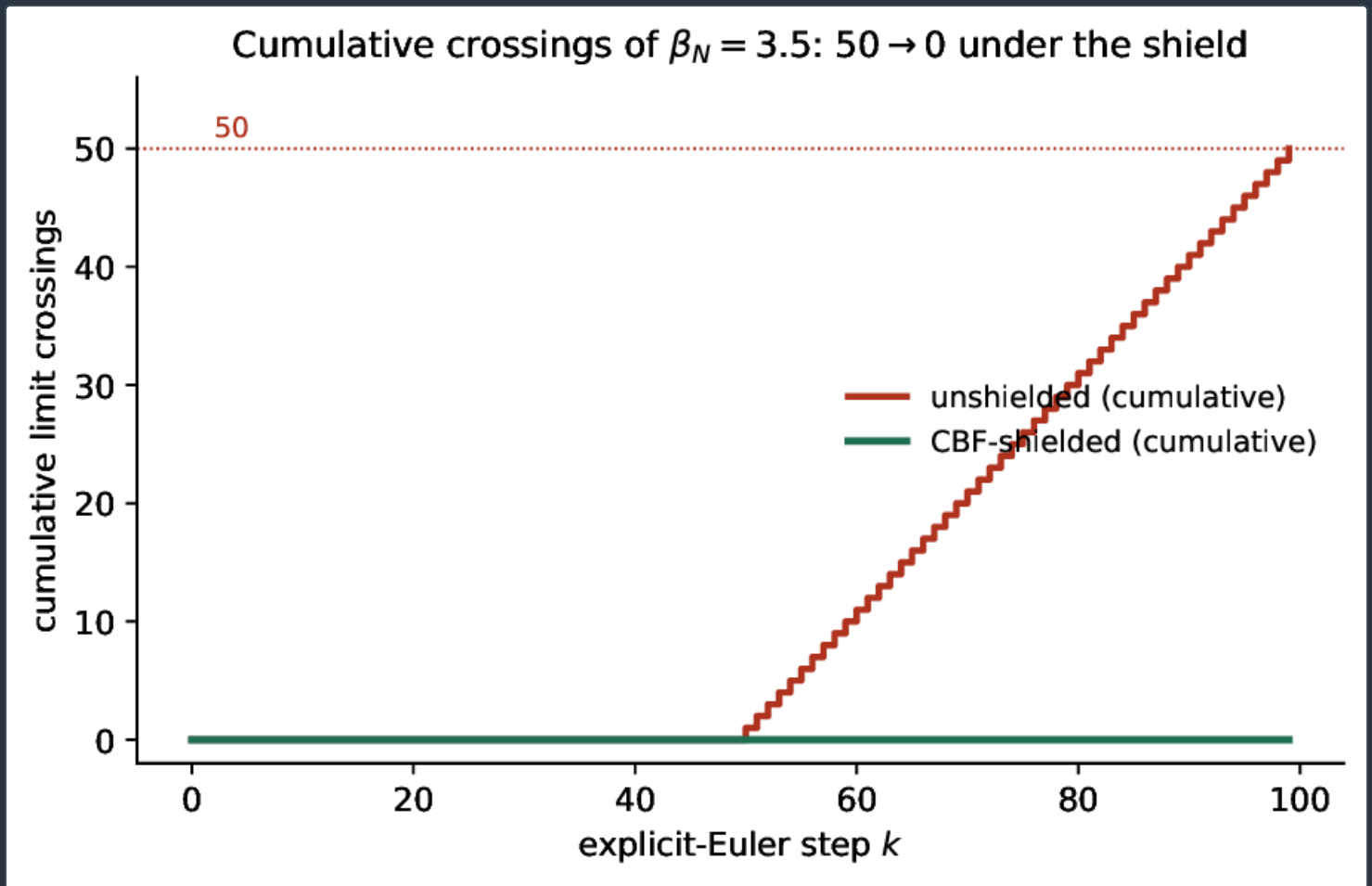
Barrier value $h(k) = 3.5 - \beta_N(k)$ along the rollout of Fig. 2. Under the shield $h \geq 0$ for all k with $\min_k h = +0.000$ (the state is held on the boundary of the safe set but never outside it); the unshielded loop drives h negative. Non-negativity of h is forward invariance made visible.

BOUNDED, NON-SATURATING AUTHORITY

The certificate is meaningful only if the actuator authority it demands is available. Figure 4 shows the realised clamp command: it stays within $u \in [1.061, 1.250]$ and never reaches the available floor $u_{\min} = 0.5$, so the barrier certificate is *not* authority-limited on this sequence — the shield holds the boundary with margin to spare. Figure 5 shows the cumulative limit crossings accumulating to **50** for the unshielded loop while the shielded loop remains flat at **0**: the two curves are the whole safety story in one frame.



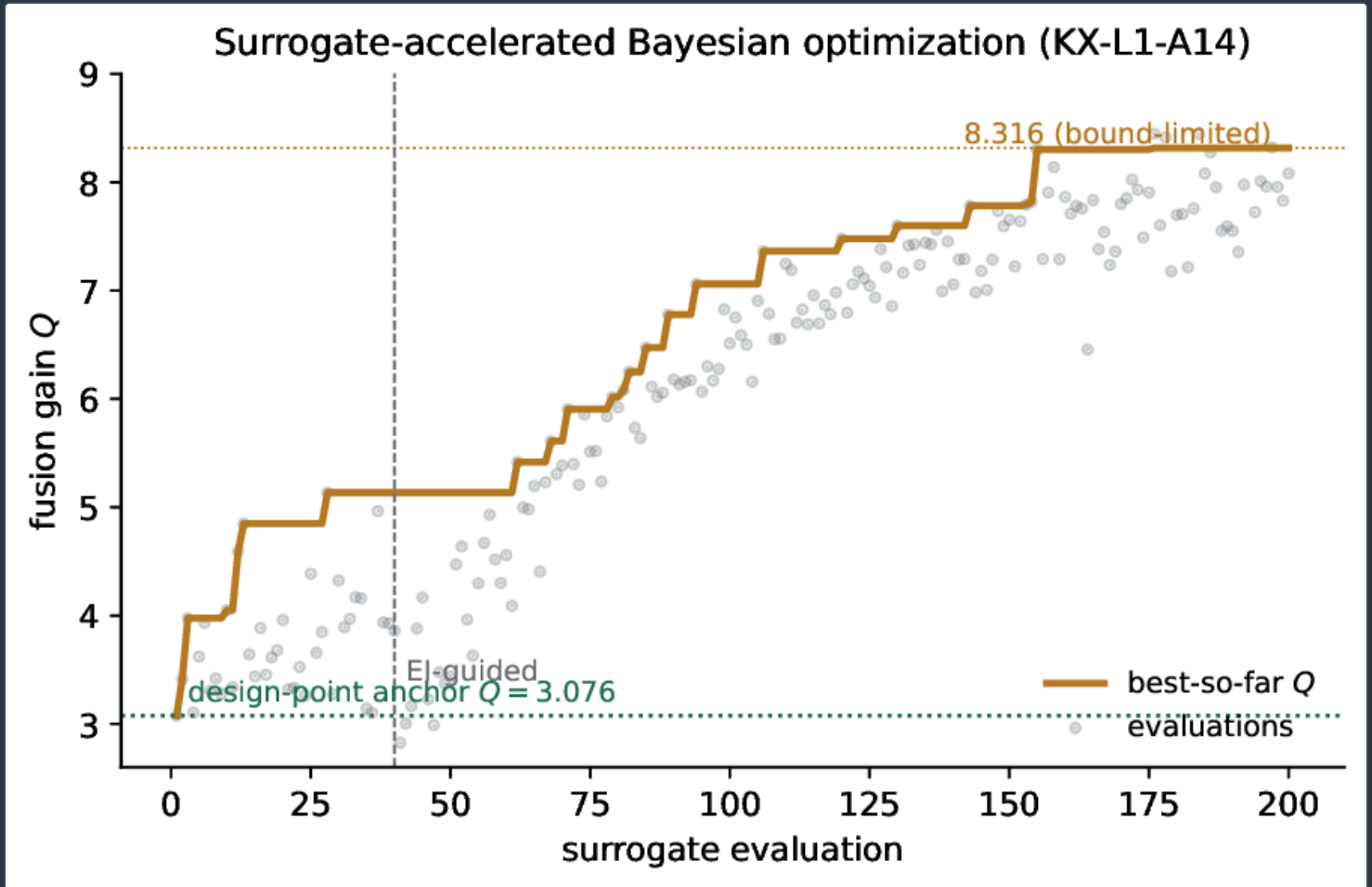
Realised clamp actuation $u(k)$ under the shield. The command stays inside $[1.061, 1.250]$ (dotted) and never reaches the available authority floor $u_{\min} = 0.5$ (dashed red); the certificate is therefore not authority-limited on this sequence — there is spare actuation between the realised command and the floor.



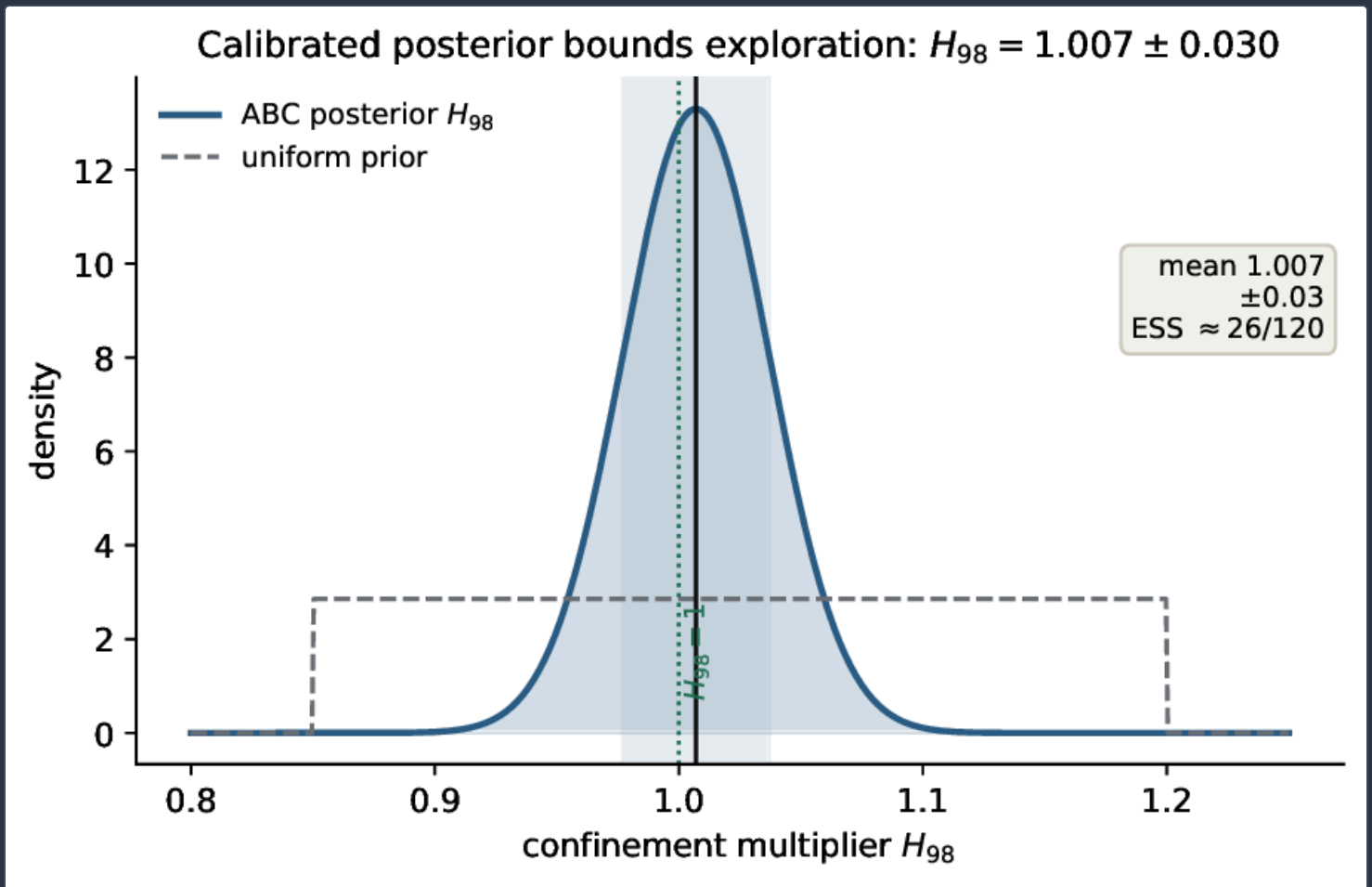
Cumulative crossings of the certified limit $\beta_N = 3.5$ along the campaign. The unshielded loop accumulates **50** crossings; the CBF-shielded loop accumulates none. The **50 $\rightarrow$ 0** reduction is the demonstrated safety guarantee.

CALIBRATED BOUND ON EXPLORATION

How far the policy may explore is bounded by a calibrated confinement posterior. The surrogate Bayesian-optimisation campaign (anchor KX-L1-A14) reproduces the design-point anchor exactly ($Q = 3.076$, $P_{\text{fus}} = 85.04 \text{ MW}$) and, over 200 evaluations, locates a best-so-far $Q = 8.316$ that sits on the prior bounds (Fig. 6) — a *bound-limited artefact*, not a new design point, and reported as such. The value of the campaign for safe RL is not that apparent optimum but the ABC posterior it yields on the confinement multiplier, $H_{98} = 1.007 \pm 0.030$ (effective sample size ≈ 26 of 120; Fig. 7). That calibrated ± 0.030 spread is what the safety layer reads to decide how far the learned policy may extrapolate before its authority is throttled ^[43,44]: a tight, well-centred posterior on $H_{98} \approx 1$ is the quantitative license for a modest exploration radius around the validated envelope.



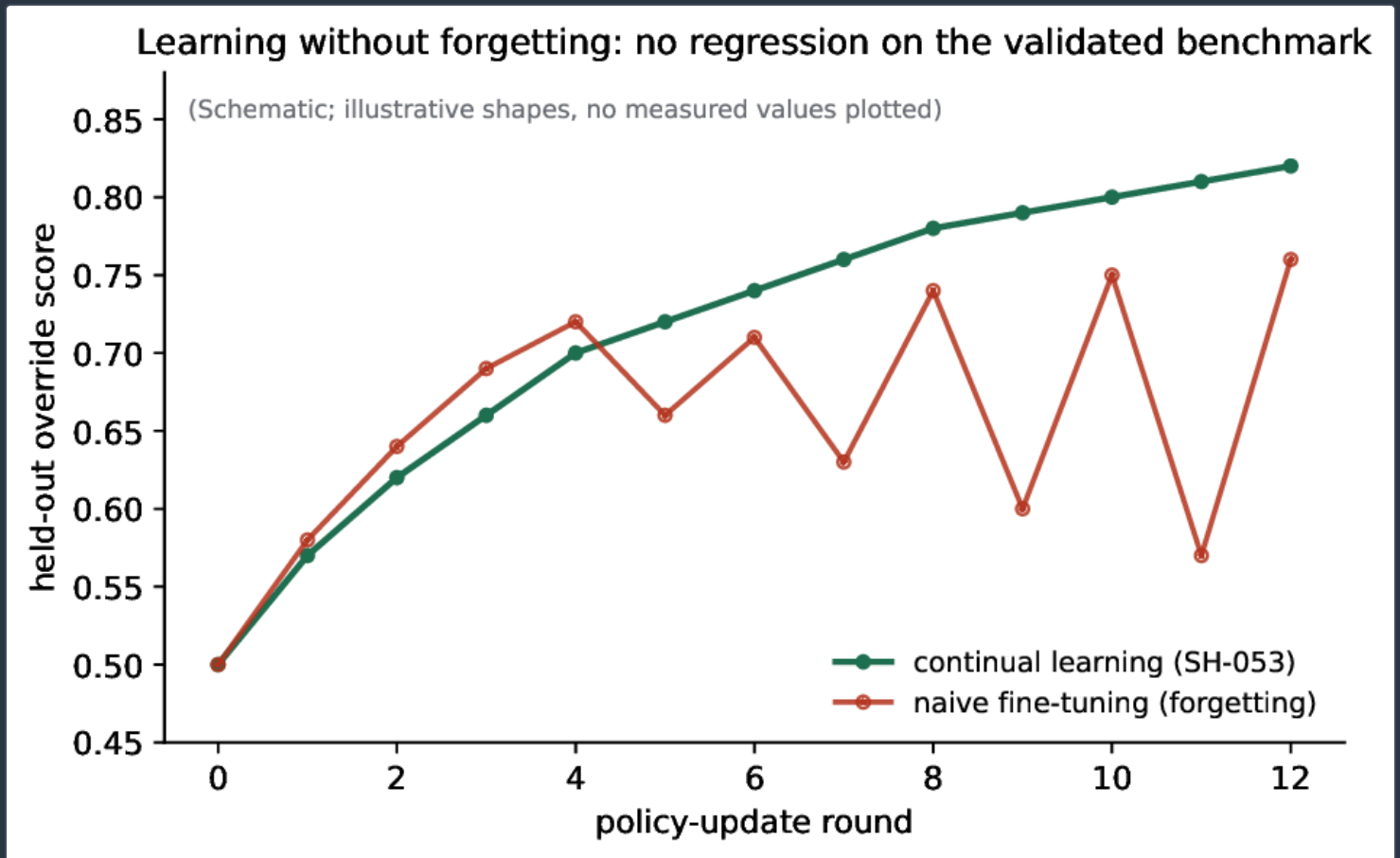
Surrogate-accelerated Bayesian optimisation (anchor KX-L1-A14): best-so-far fusion gain Q over 200 evaluations (40 initial including the design-point anchor, then expected-improvement-guided). The design-point anchor $Q = 3.076$ is reproduced; the best-so-far climbs to $Q = 8.316$ but rests on the prior bounds and is therefore a bound-limited artefact, not a certified operating point.



Calibrated confinement posterior from ABC (anchor KX-L1-A14). Given $Q_{\text{obs}} = 3.076$ at tolerance $\sigma = 5\%$, the posterior on the confinement multiplier is $H_{98} = 1.007 \pm 0.030$ (effective sample size ≈ 26 of 120), recovering $H_{98} \approx 1$ as required for self-consistency. This calibrated spread bounds how far the safe-RL policy may explore beyond the validated envelope.

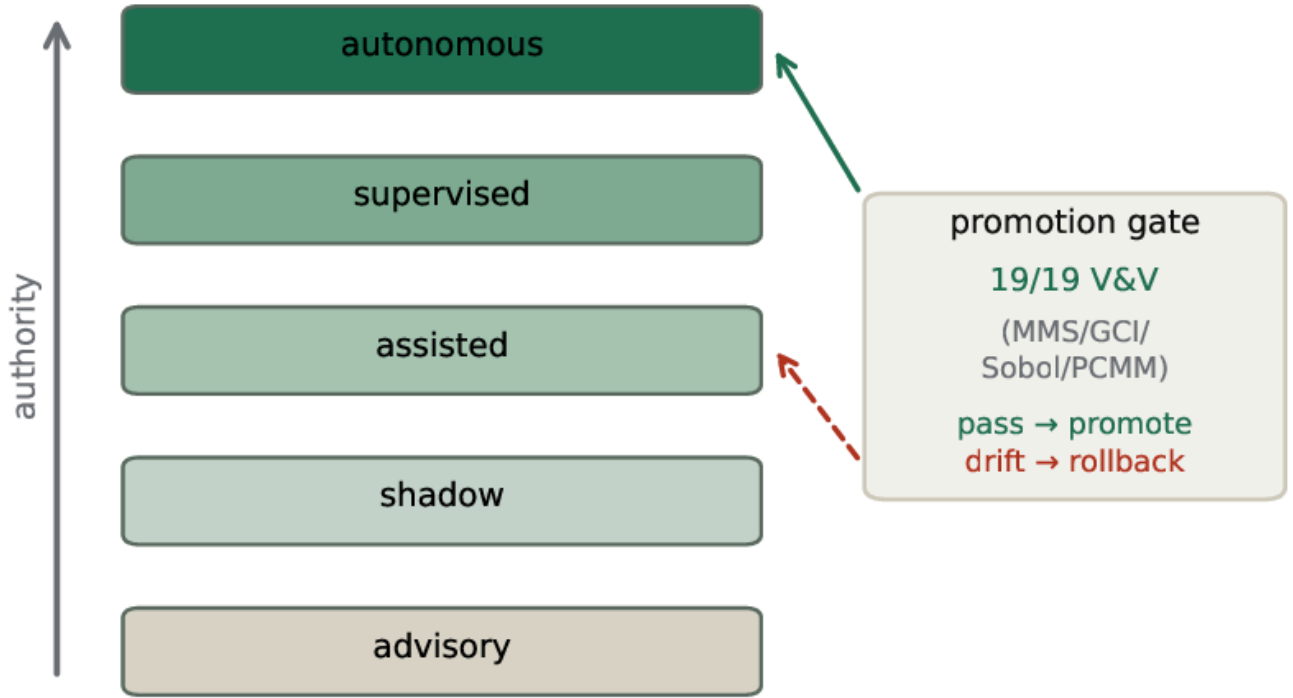
LEARNING WITHOUT FORGETTING, AND GATED PROMOTION

Figure 8 contrasts, schematically, a continual-learning update (15) that improves held-out override performance monotonically with a naive fine-tuning update that improves then regresses (catastrophic forgetting); the acceptance test admits only the former. Figure 9 shows the maturity-gated authority ladder (16): a model climbs from advisory to autonomous authority only as it clears verification maturity, and the promotion gate admits only **19/19**-passing models, rolling back on injected drift. The learned policy, however competent, never earns authority it has not verified, and never acts outside the filter.



(Schematic.) Continual learning without forgetting (serial KFE-CF-SH-053, eq. (15)) improves held-out override performance monotonically across update rounds; naive fine-tuning improves then regresses. The acceptance criterion is no regression on the validated benchmark. Shapes are illustrative; no measured values are plotted.

Authority capped by verification maturity; only 19/19-passing models promote



(Schematic.) Maturity-gated authority (serials KFE-CF-SH-054/SH-053, eq. (16)). Runtime authority climbs from advisory to autonomous only as verification maturity accrues; the promotion gate admits only models clearing the **19/19** V&V suite (anchor KX-L1-A6, MMS/GCI/Sobol/PCMM) and rolls back on drift.

ACCEPTANCE CRITERIA AS INEQUALITIES

No model is granted authority by assertion; each clears the following pre-registered inequalities, collected here as the contract that gates the loop:

(A1) forward invariance	$h(\backslash \text{bm} x_k) \geq 0 \ \forall k, \quad (\text{crossings} = 0/100)$	(KX-L1-A13, eqs.-???,???)
(A2) bounded authority	$u_k \in [u_{\min}, u_{\max}], \ u_k \neq u^{\text{floor}}$	(KX-L1-A13)
(A3) learn on overrides	improve on held-out $\mathcal{D}_{\text{test}}$ inside $\mathcal{K}_{\text{chf}}$	(SH-050, eq.-???)
(A4) informative queries	$\mathbb{I}[\Theta; y]_{\text{design}} > \mathbb{I}[\Theta; y]_{\text{random}}$	(SH-052, eq.-???)
(A5) no forgetting	$\Delta(\text{validated benchmark}) \geq 0$ after update	(SH-053, eq.-???)
(A6) gated promotion	$\text{promote} \iff m = 1 \text{ (19/19), else rollback}$	(SH-054/A6, eq.-???)
(A7) calibrated bound	$H_{98} = 1.007 \pm 0.030$ caps exploration radius	(KX-L1-A14)

Authority is capped by verification maturity until each bar is met.

Discussion

WHY SEPARATING WHERE

from *how* is what makes RL admissible Prior learned tokamak controllers achieved their guarantees, where they had them, by careful reward design and extensive in-simulation training^[1,2]; the safety argument was empirical — the policy did not misbehave in testing. That is a weaker claim than forward invariance. By contrast, the design here makes the safety argument *structural*: because every executed action is the projection $\Pi_{\mathcal{C}}(\pi_{\theta})$ onto the admissible set (7), the closed loop cannot leave $\mathcal{C}$ for *any* θ , including a half-trained or adversarially perturbed one. The demonstrated $50 \rightarrow 0$ reduction in limit crossings, with $\min_k h = +0.000$, is the empirical footprint of that structural guarantee. This is the safe-RL-by-shielding paradigm^[15,17] instantiated on the specific limit — pressure-limited β_N — that most directly threatens a compact, high- β spherical-tokamak breeder.

AGAINST THE SAFE-RL AND CONSTRAINED-CONTROL LITERATURE

Three families of safe-RL methods bear comparison. Lagrangian / constrained-policy methods (CPO and relatives^[14]) bound *expected* constraint cost; they do not preclude the rare catastrophic excursion that eq. (5) forbids, which is why we enforce an almost-sure state constraint by a filter rather than a budget in the objective. Reward-shaping approaches^[16] inherit the same weakness. Shielding^[15] and CBF-QP filtering^[17,18,19] provide the during-learning guarantee we require — as do the reachability-based and predictive safety filters that share its minimal-intervention structure^[20,21]; our contribution is to fuse that filter with an operator-feedback learning signal (eqs. 10–12) and a calibrated exploration bound, and to demonstrate the composite on a reproduced fusion-control anchor. Relative to offline model-based RL for tokamak control^[6], which learns entirely from logged data without any online exploration, our loop *does* explore — but only inside the certified set — and thereby retains RL’s ability to discover strategies the logged data does not contain, without giving up the safety floor.

OPERATOR FEEDBACK AS THE RIGHT DATA SOURCE

Learning from the operator logbook rather than from unconstrained trial has a second virtue beyond safety: the policy improves along the trajectories operators actually use, which is where its competence is most valuable and where the twin surrogates are best validated. The override reward (11), grounded by twin replay rather than by preference alone, converts each expert correction into a physically scored lesson — the reinforcement-learning-from-human-feedback recipe^[11,12] adapted to a setting where the “human preference” can be checked against physics before it is trusted. The active-learning designer (14) then spends the scarce budget of informative queries where they most reduce posterior uncertainty^[24,48], and the continual-learning penalty (15) ensures each such lesson is retained without erasing an older one.

Limitations

One honest gate remains. The override reward (11) is at present shaped from twin-synthetic operator corrections; grounding it in the accumulating *live* operator logbook is the one confirming step, and it is the natural first use of operating experience as it accrues. Separately, the demonstrated β_N dynamics are a one-dimensional, design-point-anchored surrogate: the forward-invariance certificate is proven at that reduced level (over **100** explicit-Euler steps), and its extension to the full coupled equilibrium–transport plant is the subject of the reference-governor MPC and four-twin architecture developed in the companion papers ^[42,41]. The safety shield that makes exploration safe to run is demonstrated now, independent of both steps.

Conclusion

Kronos safe reinforcement learning improves the controller from operator feedback while confining every exploratory action to a control-barrier-certified safe set. The design decouples the two questions safe learning usually conflates: *where* the policy may act is decided once, by a CBF-certified set that no action may leave, and *how* it improves inside that set is decided by learning from the operators — imitation from a hash-chained logbook and a twin-scored override reward, disciplined by an active-learning designer that beats random selection, continual learning without forgetting, and gated promotion with rollback. The safety guarantee is demonstrated, not promised: zero limit crossings in a 100-step exploration campaign against 50 unshielded, forward invariance verified with $\min_k h = +0.000$, and a bounded actuation $u \in [1.061, 1.250]$ that never reaches the $u_{\min} = 0.5$ floor, all bounded by a calibrated $H_{98} = 1.007 \pm 0.030$ posterior. The controller gets better from the people who run it without ever leaving the set that is proven safe.

Acknowledgments Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos physics de-risking programme for the frozen, independently reproduced result set underlying every quantitative claim, and the KRONOS-CTRL computing-and-control effort for the codes and the NVIDIA GPU HPC campaign.

Reproducibility. The safe-learning anchors are frozen entries in the Kronos Physics De-Risking Register ^[49] ([doi:10.5281/zenodo.22133057](https://doi.org/10.5281/zenodo.22133057)): the CBF shield (KX-L1-A13: $\beta_N 4.200 \rightarrow 3.500$, $50 \rightarrow 0$ crossings, $\min_k h = +0.000$, $u \in [1.061, 1.250]$), the surrogate Bayesian-optimisation calibration (KX-L1-A14: $H_{98} = 1.007 \pm 0.030$), and the formal V&V suite (KX-L1-A6: 19/19). Each is regenerable from the archived closed-loop, UQ and verification runs under a single global seed (20260726), and every figure is regenerated by the deposited script `make_fig_2p81_safe_rl.py` from these anchors. This paper is archived at its DOI [doi:10.5281/zenodo.22645823](https://doi.org/10.5281/zenodo.22645823) and its official page kronosfusionenergy.com/publications/safe-reinforcement-learning-from-operator-feedback-for-.

Data availability This paper is archived at [doi:10.5281/zenodo.22645823](https://doi.org/10.5281/zenodo.22645823). The shield definition, the operator-feedback learning objectives, the operator-logbook schema and the figure-generation script are archived with the deposit and reference the Kronos 2026 series, including the AI and quantum-control overview ^[50] ([doi:10.5281/zenodo.22645702](https://doi.org/10.5281/zenodo.22645702), [official page](#)). All quantitative values trace to the register ^[49] by the serial and anchor identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests Provisional patent applications relating to aspects of this work (KRONOS-CTRL) have been filed.

References

A P P A R A T U S

Portfolio, People & Record

*The intellectual-property estate, the people who built it, the limitations
that gate it, and the sources it rests on.*

1

GRANTED PATENT

4

2026 PROVISIONALS

40

TEAM MEMBERS

27

2022 PROVISIONALS

INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

US 12,009,112	High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550	GRANTED
US 17/878,507	Core device / method (KRONO-001A) · utility	PENDING

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

64/124,209	Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026	FILED
64/124,220	Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026	FILED
64/115,167	KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026	FILED
64/005,440	Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026	FILED
64/105,530	Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026	FILED
64/128,097	Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop	FILED

TRADEMARK & ORIGIN

FTK-98801894	Kronos Fusion Energy — trademark application	FILED
KR-2022-★	Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022	PRIORITY

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any $Q > 1$ milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

THE TEAM

Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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Andrea Romero

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Legal counsel is retained; those roles are held on the internal roster and are not listed here.

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